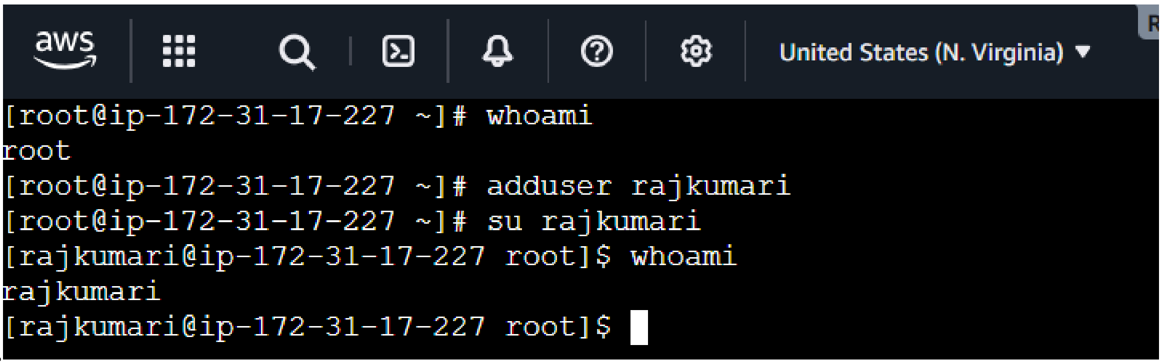
Command : whoami

Objective -> To create a new user account and switch from the root user to the newly created user in Linux

Purpose -> The whoami command displays the username of the currently logged-in user.

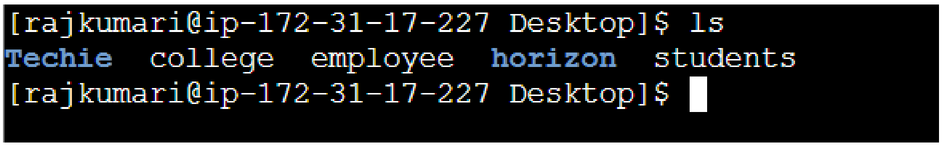



```
[root@ip-172-31-17-227 ~]# whoami
root
[root@ip-172-31-17-227 ~]# adduser rajkumari
[root@ip-172-31-17-227 ~]# su rajkumari
[rajkumari@ip-172-31-17-227 root]$ whoami
rajkumari
[rajkumari@ip-172-31-17-227 root]$
```

Cmd: ls

Objective -> To display the files and directories present in the current working directory.

Purpose –> The ls command is used to list the contents of a directory. It helps users view available files and folders for navigation and file management.

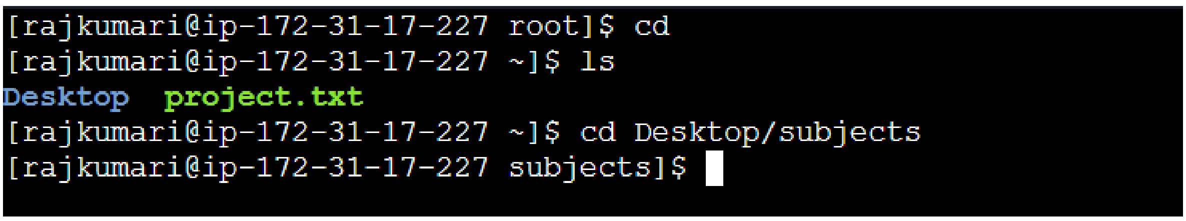


```
[rajkumari@ip-172-31-17-227 Desktop]$ ls
Techie  college  employee  horizon  students
[rajkumari@ip-172-31-17-227 Desktop]$
```

Cmd : cd

Objective -> To change the current working directory.

Purpose -> The cd command is used to navigate between different directories in the Linux file system.



```
[rajkumari@ip-172-31-17-227 root]$ cd
[rajkumari@ip-172-31-17-227 ~]$ ls
Desktop  project.txt
[rajkumari@ip-172-31-17-227 ~]$ cd Desktop/subjects
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: ip a

Objective ->To display the IP address and network interface information of the Linux system.

Purpose -> The ip a command is used to view all network interfaces and their IP addresses. It helps administrators verify network configuration, troubleshoot connectivity issues, and identify interface status.

```
[rajkumari@ip-172-31-17-227 Desktop]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
len 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state U
 qlen 1000
    link/ether 0a:ff:e5:e1:35:8f brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-0bb96eda2c7737357
    altname device-number-0.0
    inet 172.31.17.227/20 metric 512 brd 172.31.31.255 scope global
       valid_lft 2510sec preferred_lft 2510sec
    inet6 fe80::8ff:e5ff:fee1:358f/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
```

Cmd: cat

Objective -> To display, create, combine, and manage the contents of text files in Linux.

Purpose -> The cat (concatenate) command is used to view the contents of one or more text files. It can also be used to create new files, merge multiple files, and append content to existing files.

```
[rajkumari@ip-172-31-17-227 Desktop]$ cat raj.txt
I am very good student
[rajkumari@ip-172-31-17-227 Desktop]$
```

Cmd: touch

Objective -> To create a new empty file.

Purpose -> The touch command creates a new file or updates the timestamp of an existing file.

```
[rajkumari@ip-172-31-17-227 Desktop]$ touch hotel.txt
[rajkumari@ip-172-31-17-227 Desktop]$ ls
Techie  college  employee  horizon  hotel.txt  raj.txt  students
[rajkumari@ip-172-31-17-227 Desktop]$
```

Cmd: pwd

Objective -> To display the current working directory.

Purpose -> The pwd command shows the complete path of the current directory, helping users know their current location in the file system.

```
[rajkumari@ip-172-31-17-227 Desktop]$ pwd
/home/rajkumari/Desktop
[rajkumari@ip-172-31-17-227 Desktop]$ 
```

Cmd: man

Objective -> To access the Linux manual pages and learn about commands, utilities, and system functions.

Purpose -> The man (manual) command displays detailed documentation about Linux commands, including their syntax, options, description, and examples. It is the primary help system used by Linux administrators and users.

```
[rajkumari@ip-172-31-17-227 Desktop]$ man
What manual page do you want?
For example, try 'man man'.
[rajkumari@ip-172-31-17-227 Desktop]$ man ls
[rajkumari@ip-172-31-17-227 Desktop]$ 
```

```
LS(1)                                                                  User Co
mmands                                                                 LS(1)
NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              do not ignore entries starting with .

       -A, --almost-all
              do not list implied . and ..

       --author
              with -l, print the author of each file

       -b, --escape
              print C-style escapes for nongraphic characters
              do not list implied entries ending with ~

       -c     with  -lt:  sort  by, and show, ctime (time of last modification of f
ile status information); with -l: show ctime and sort by name; otherwise: sort by
              ctime, newest first
       -c     with  -lt:  sort  by, and show, ctime (time of last modification of f
ile status information); with -l: show ctime and sort by name; otherwise: sort by
```

File Management Commands

Cmd: cp

Objective -> To copy files or directories.

Purpose -> The cp command duplicates files or directories from one location to another.

```
[rajkumari@ip-172-31-17-227 Desktop]$ ls
Techie  college  employee  horizon  hotel.txt  raj.txt  students
[rajkumari@ip-172-31-17-227 Desktop]$ cp raj.txt Techie
[rajkumari@ip-172-31-17-227 Desktop]$ cd Techie
[rajkumari@ip-172-31-17-227 Techie]$ ls
raj.txt
[rajkumari@ip-172-31-17-227 Techie]$
```

Cmd: mkdir

Objective -> To create a new directory.

Purpose -> The mkdir command creates one or more directories to organize files and folders.

```
[rajkumari@ip-172-31-17-227 Techie]$ mkdir Polity
[rajkumari@ip-172-31-17-227 Techie]$ ls
Polity  raj.txt
[rajkumari@ip-172-31-17-227 Techie]$ mkdir General
[rajkumari@ip-172-31-17-227 Techie]$ ls
General  Polity  raj.txt
[rajkumari@ip-172-31-17-227 Techie]$
```

Cmd: rmdir

Objective -> To remove files or directories.

Purpose -> The rm command permanently deletes files or directories from the system.

```
Techie  college  employee  horizon  hotel.txt  raj.txt  students
[rajkumari@ip-172-31-17-227 Desktop]$ rmdir horizon
[rajkumari@ip-172-31-17-227 Desktop]$ ls
Techie  college  employee  hotel.txt  raj.txt  students
[rajkumari@ip-172-31-17-227 Desktop]$
```

Cmd: mv

Objective -> To move or rename files and directories.

Purpose -> The mv command is used to move files/directories or rename them without changing their contents.

```
[rajkumari@ip-172-31-17-227 Techie]$ ls
General  Polity  raj.txt
[rajkumari@ip-172-31-17-227 Techie]$ mv General Science
[rajkumari@ip-172-31-17-227 Techie]$ ls
Polity  Science  raj.txt
[rajkumari@ip-172-31-17-227 Techie]$
```

Cmd: rm

Objective -> To remove (delete) files and directories from the Linux file system.

Purpose -> The rm (remove) command is used to permanently delete files and directories. It helps users manage disk space by removing unnecessary or unwanted files and folders.

```
[rajkumari@ip-172-31-17-227 Desktop]$ ls
Techie  employee  emplyees  hotel.txt  raj.txt  students
[rajkumari@ip-172-31-17-227 Desktop]$ rm students
[rajkumari@ip-172-31-17-227 Desktop]$ rmdir emplyees
[rajkumari@ip-172-31-17-227 Desktop]$ ls
Techie  employee  hotel.txt  raj.txt
[rajkumari@ip-172-31-17-227 Desktop]$
```

Advance File Management Commands

Cmd: ln -s

Objective -> To create a symbolic (soft) link that points to another file or directory.

Purpose -> The ln -s command is used to create a symbolic link (also called a soft link). A symbolic link acts like a shortcut to the original file or directory, making it easier to access without duplicating the data.

```
[rajkumari@ip-172-31-17-227 Desktop]$ ln -s students
[rajkumari@ip-172-31-17-227 Desktop]$ ls
Techie  employee  hotel.txt  raj.txt  students
[rajkumari@ip-172-31-17-227 Desktop]$
```

Cmd: find

Objective: To search for files and directories in the Linux file system.

Purpose: The find command is used to locate files and directories based on various criteria such as name, type, size, permissions, owner, and modification time. It is commonly used for file management, system administration, and automating tasks by performing actions like deleting, copying, or changing permissions on the search results.

```
[rajkumari@ip-172-31-17-227 Desktop]$ cd
[rajkumari@ip-172-31-17-227 ~]$ find ~/ -name raj.txt
/home/rajkumari/Desktop/Techie/raj.txt
/home/rajkumari/Desktop/raj.txt
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: tar

Objective -> To archive multiple files and directories into a single file and to extract archived files.

Purpose -> The tar (Tape Archive) command is used to create, view, extract, and manage archive files. It combines multiple files and directories into a single archive, making it easier to store, transfer, and back up data.

```
[rajkumari@ip-172-31-17-227 Desktop]$ tar -cvf archive.tar  Techie
Techie/
Techie/raj.txt
Techie/Polity/
Techie/Science/
[rajkumari@ip-172-31-17-227 Desktop]$
```

Cmd: gzip

Objective -> To compress and decompress files using the **gzip** compression utility.

Purpose -> The gzip command is used to reduce the size of files for efficient storage and faster file transfer. It compresses a single file and creates a .gz file.

Command	**Description**
gzip file.txt	Compress a file
gunzip file.txt.gz	Decompress a file
gzip -d file.txt.gz	Decompress a file
gzip -k file.txt	Keep the original file after compression

Command	Description
gzip -l file.txt.gz	Display compression information

```
[rajkumari@ip-172-31-17-227 Desktop]$ gzip hotel.txt
[rajkumari@ip-172-31-17-227 Desktop]$ ls
Techie  archive.tar  employee  hotel.txt.gz  raj.txt  students
[rajkumari@ip-172-31-17-227 Desktop]$
```

Cmd: bzip2

Objective -> To compress and decompress files using the **bzip2** compression utility.

Purpose -> The bzip2 command compresses files more efficiently than gzip, resulting in smaller file sizes, although it takes longer to compress.

```
[rajkumari@ip-172-31-17-227 Desktop]$ bzip2 english.txt
[rajkumari@ip-172-31-17-227 Desktop]$ ls
Techie  archive.tar  employee  english.txt.bz2  hotel.txt.gz  maths.txt  raj.txt  s
tudents
[rajkumari@ip-172-31-17-227 Desktop]$
```

Cmd: xz

Objective -> To compress and decompress files using the **XZ** compression algorithm.

Purpose -> The xz command offers a very high compression ratio, making it suitable for large files and software packages.

```
[rajkumari@ip-172-31-17-227 subjects]$ ls
english  hindi  telugu
[rajkumari@ip-172-31-17-227 subjects]$ xz hindi
[rajkumari@ip-172-31-17-227 subjects]$ ls
english  hindi.xz  telugu
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: zip

Objective -> To compress one or more files and directories into a ZIP archive.

Purpose -> The zip command creates a .zip archive, which is widely supported across Linux, Windows, and macOS for file sharing.

```
[rajkumari@ip-172-31-17-227 subjects]$ ls
archive.zip  english  hindi.xz  physics  social  telugu
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: file

Objective -> To identify the type of a file in Linux.

Purpose -> The file command is used to determine the type of a file based on its contents rather than its file extension. It helps users identify text files, executable files, images, archives, scripts, and other file formats.

```
[rajkumari@ip-172-31-17-227 subjects]$ file social
social: empty
[rajkumari@ip-172-31-17-227 subjects]$ file hindi
hindi: cannot open `hindi' (No such file or directory)
[rajkumari@ip-172-31-17-227 subjects]$ file hindi.xz
hindi.xz: XZ compressed data
[rajkumari@ip-172-31-17-227 subjects]$ file archive.zip
archive.zip: Zip archive data, at least v1.0 to extract
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: mount

Objective - > To mount a storage device or file system so it becomes accessible through the Linux file system.

Purpose -> The mount command is used to attach storage devices (such as hard disks, USB drives, or ISO images) to a directory called a **mount point**, allowing users to access the files stored on the device.

Cmd: df -h

Objective -> To display disk space usage in a human-readable format.

Purpose -> The df -h command is used to check total, used, and available disk space on mounted file systems.

```
[rajkumari@ip-172-31-17-227 subjects]$ df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          420M     0  420M   0% /dev
tmpfs             457M     0  457M   0% /dev/shm
tmpfs             183M  444K  183M   1% /run
efivarfs          128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1    8.0G  1.8G  6.3G  22% /
tmpfs             457M     0  457M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: findmnt

Objective -> To display information about mounted file systems in a structured format.

Purpose -> The findmnt command lists all mounted file systems, their source devices, mount points, and file system types in a tree-like format.

```
[rajkumari@ip-172-31-17-227 subjects]$ findmnt
TARGET                                                SOURCE          FSTYPE     OPTIONS
/                                                     /dev/nvme0n1p1 xfs        rw,noatim
e,seclabel,inode64,logbufs=8,logbsize=32k,sunit=1024,swidth=1024,noquota
├─/proc                                               proc            proc       rw,nosuid
,nodev,noexec,relatime
│ └─/proc/sys/fs/binfmt_misc                          systemd-1       autofs     rw,relati
me,fd=33,pgrp=1,timeout=0,minproto=5,maxproto=5,direct,pipe_ino=1734
├─/sys                                                sysfs           sysfs      rw,nosuid
,nodev,noexec,relatime,seclabel
│ ├─/sys/kernel/security                              securityfs      securityfs rw,nosuid
,nodev,noexec,relatime
│ ├─/sys/fs/cgroup                                    cgroup2         cgroup2    rw,nosuid
,nodev,noexec,relatime,seclabel,nsdelegate,memory_recursiveprot
│ ├─/sys/fs/pstore                                    none            pstore     rw,nosuid
,nodev,noexec,relatime,seclabel
│ ├─/sys/firmware/efi/efivars                         efivarfs        efivarfs   rw,nosuid
,nodev,noexec,relatime
│ ├─/sys/fs/bpf                                       bpf             bpf        rw,nosuid
,nodev,noexec,relatime,mode=700
│ ├─/sys/fs/selinux                                   selinuxfs       selinuxfs  rw,nosuid
,noexec,relatime
│ ├─/sys/kernel/tracing                               tracefs         tracefs    rw,nosuid
,nodev,noexec,relatime,seclabel
│ ├─/sys/kernel/debug                                 debugfs         debugfs    rw,nosuid
,nodev,noexec,relatime,seclabel
│ ├─/sys/kernel/config                                configfs        configfs   rw,nosuid
,nodev,noexec,relatime
│ └─/sys/fs/fuse/connections                          fusectl         fusectl    rw,nosuid
```

Cmd: lsblk

Objective -> To display information about block storage devices connected to the Linux system.

Purpose -> The lsblk command is used to list all available block devices such as hard disks, SSDs, USB drives, and their partitions. It helps administrators identify storage devices and mount points.

```
[rajkumari@ip-172-31-17-227 subjects]$ lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0    8G  0 disk
├─nvme0n1p1   259:1    0    8G  0 part /
├─nvme0n1p127 259:2    0    1M  0 part
└─nvme0n1p128 259:3    0   10M  0 part /boot/efi
[rajkumari@ip-172-31-17-227 subjects]$
```

Working with Text file Commands

Cmd: nano editor

Objective -> To create and edit text files using the Nano text editor.

Purpose -> Nano is a simple command-line text editor suitable for beginners.

```
[rajkumari@ip-172-31-17-227 subjects]$ nano file1.txt
[rajkumari@ip-172-31-17-227 subjects]$ ls
archive.zip  english  file1.txt  hindi.xz  physics  social  telugu
[rajkumari@ip-172-31-17-227 subjects]$ cat file1.txt
Welcome to Blue Moon

[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: vim

Objective -> To create and edit text files using the Vim editor.

Purpose -> Vim is a powerful text editor widely used by Linux administrators and developers.

```
[rajkumari@ip-172-31-17-227 subjects]$ ls
archive.zip  english  file1.txt  file2.txt  hindi.xz  physics  social  telugu
[rajkumari@ip-172-31-17-227 subjects]$ vim file3.txt
[rajkumari@ip-172-31-17-227 subjects]$ cat file3.txt
have a good dayyyyyyyyyyyyyyyyyy
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: more

Objective -> To view large text files one screen at a time.

Purpose -> The more command allows users to read large files page by page

```
[rajkumari@ip-172-31-17-227 subjects]$ more file1.txt
Welcome to Blue Moon
```

Cmd: head

Objective -> To display the first few lines of a file.

Purpose -> The head command helps users quickly view the beginning of a file.

```
[rajkumari@ip-172-31-17-227 subjects]$ head -4 file4.txt
"Lunareth — where moonlight meets strength, and dreams become destiny."
"Like Lunareth, shine quietly; the brightest light doesn't always make the loudest
noise."
"In the silence of the night, Lunareth reminds us that even darkness has its own li
ght."
"Lunareth is not just a name—it's a symbol of hope that glows even in the darkest s
kies."
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: tail

Objective -> To display the first few lines of a file.

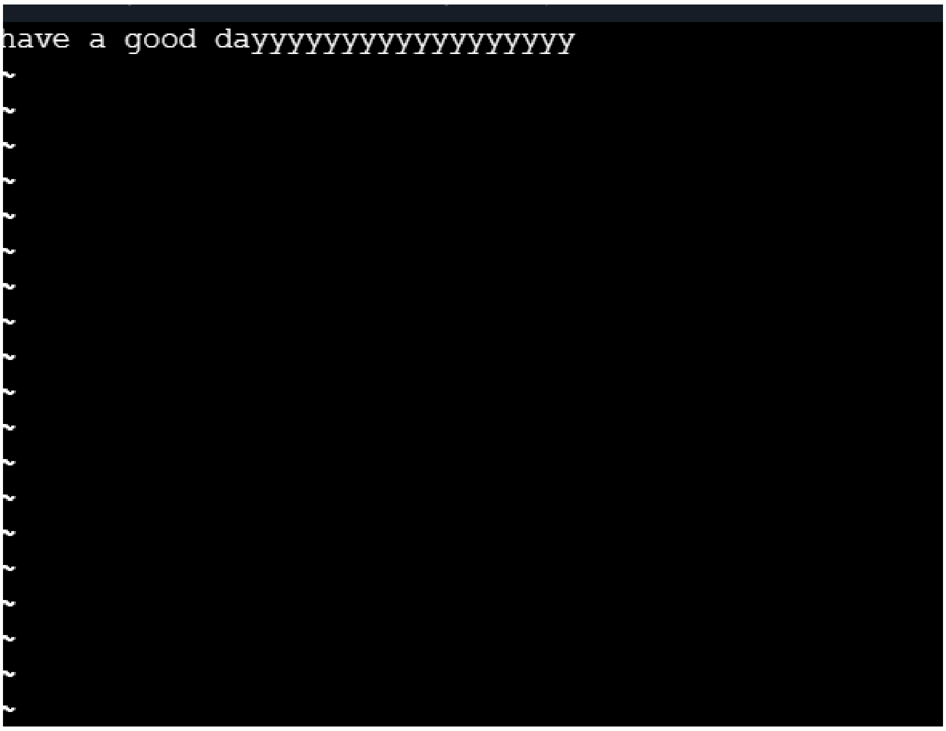
Purpose -> The head command helps users quickly view the beginning of a file.

```
[rajkumari@ip-172-31-17-227 subjects]$ tail -2 file4.txt
"The moon doesn't compete with the sun; it shines in its own time. Be Lunareth."
"Lunareth is the quiet promise that even the darkest night ends in light."
[rajkumari@ip-172-31-17-227 subjects]$
```

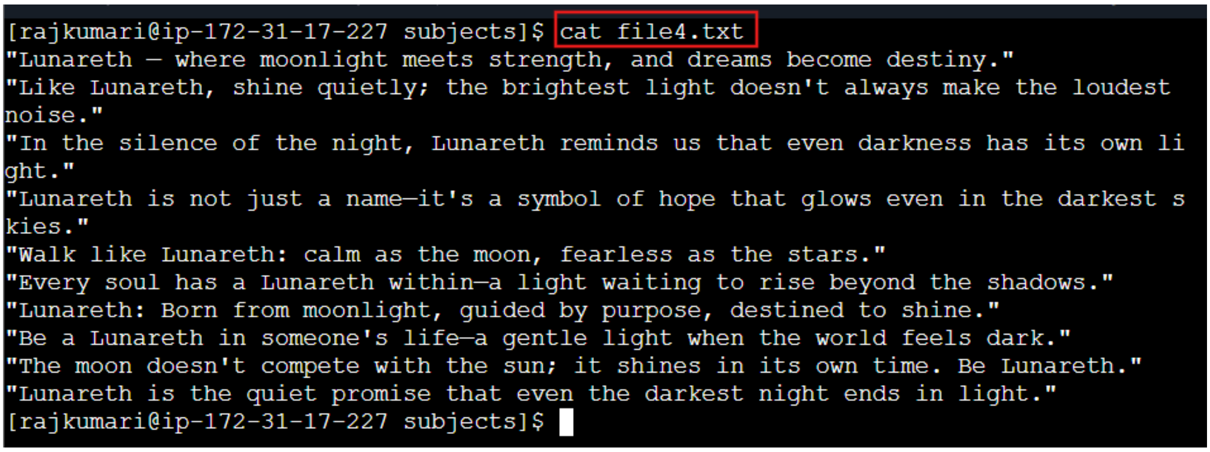
Cmd: less

Objective -> To view large text files with forward and backward navigation.

Purpose -> The less command displays large files one screen at a time and allows users to move both forward and backward.



Cmd: cat



Cmd: cat -A

Objective -> To display the contents of a file along with special characters such as tabs, end-of-line markers, and non-printable characters.

Purpose -> The cat -A command is used to view hidden characters in a text file. It is useful for debugging formatting issues and identifying tabs, spaces, and line endings.

```
[rajkumari@ip-172-31-17-227 subjects]$ cat -A file4.txt
"Lunareth M-bM-^@M-^T where moonlight meets strength, and dreams become destiny."$
"Like Lunareth, shine quietly; the brightest light doesn't always make the loudest
noise."$
"In the silence of the night, Lunareth reminds us that even darkness has its own li
ght."$
"Lunareth is not just a nameM-bM-^@M-^Tit's a symbol of hope that glows even in the
 darkest skies."$
"Walk like Lunareth: calm as the moon, fearless as the stars."$
"Every soul has a Lunareth withinM-bM-^@M-^Ta light waiting to rise beyond the shad
ows."$
"Lunareth: Born from moonlight, guided by purpose, destined to shine."$
"Be a Lunareth in someone's lifeM-bM-^@M-^Ta gentle light when the world feels dark
."$
"The moon doesn't compete with the sun; it shines in its own time. Be Lunareth."$
"Lunareth is the quiet promise that even the darkest night ends in light."$
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: tac

Objective -> To display the contents of a file in reverse line order.

Purpose -> The tac command prints the last line first and the first line last. It is the reverse of the cat command.

```
[rajkumari@ip-172-31-17-227 subjects]$ tac file4.txt
"Lunareth is the quiet promise that even the darkest night ends in light."
"The moon doesn't compete with the sun; it shines in its own time. Be Lunareth."
"Be a Lunareth in someone's life—a gentle light when the world feels dark."
"Lunareth: Born from moonlight, guided by purpose, destined to shine."
"Every soul has a Lunareth within—a light waiting to rise beyond the shadows."
"Walk like Lunareth: calm as the moon, fearless as the stars."
"Lunareth is not just a name—it's a symbol of hope that glows even in the darkest s
kies."
"In the silence of the night, Lunareth reminds us that even darkness has its own li
ght."
"Like Lunareth, shine quietly; the brightest light doesn't always make the loudest
noise."
"Lunareth — where moonlight meets strength, and dreams become destiny."
[rajkumari@ip-172-31-17-227 subjects]$
```

Advances Text File Processing Commands

Cmd: grep

Objective -> To search for specific text or patterns in files.

Purpose -> The grep command searches files for matching text and displays the matching lines.

```
[rajkumari@ip-172-31-17-227 subjects]$ grep moon file4.txt
"Lunareth — where moonlight meets strength, and dreams become destiny."
"Walk like Lunareth: calm as the moon, fearless as the stars."
"Lunareth: Born from moonlight, guided by purpose, destined to shine."
"The moon doesn't compete with the sun; it shines in its own time. Be Lunareth."
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: awk

Objective -> To process and extract specific fields from text files.

Purpose -> The awk command is used for pattern scanning and text processing. It is commonly used to print columns from structured text.

```
[rajkumari@ip-172-31-17-227 subjects]$ cat marks.txt
raj 80
sita 90
ram 70
ravi 60
geetha 100
[rajkumari@ip-172-31-17-227 subjects]$ awk '{sum += $2} END {print "Average =" , su
m/NR}' marks.txt
Average = 80
[rajkumari@ip-172-31-17-227 subjects]$ awk '{sum += $2} END {print "Sum=" , sum}' m
arks.txt
Sum= 400
[rajkumari@ip-172-31-17-227 subjects]$ awk 'END {print "count =" ,NR}' marks.txt
count = 5
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: sed

Objective -> To search, replace, insert, and delete text in a file or stream.

Purpose -> The sed (Stream Editor) command edits text automatically without opening the file in an editor.

```
[rajkumari@ip-172-31-17-227 subjects]$ vi employ
[rajkumari@ip-172-31-17-227 subjects]$ cat employ
101 raj manager
102 prias developer
103 anil manager
104 kiran tester
[rajkumari@ip-172-31-17-227 subjects]$ sed 's/manager/teamlead/' employ
101 raj teamlead
102 prias developer
103 anil teamlead
104 kiran tester
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: cut

Objective -> To extract specific columns or fields from a file.

Purpose -> The cut command extracts selected portions of each line from a file based on characters, bytes, or field delimiters.

```
[rajkumari@ip-172-31-17-227 subjects]$ cut -c1-3 file5.txt
raj
lin
apa
dev
eng
[rajkumari@ip-172-31-17-227 subjects]$ cut -c4-8 file5.txt
kumar
ux
cheto
ops
ineer
```

Cmd: tr

Objective -> To translate, replace, or delete characters from the input text.

Purpose -> The tr (translate) command is used to replace one set of characters with another, convert lowercase letters to uppercase (or vice versa), remove unwanted characters, and compress repeated characters.

```
[rajkumari@ip-172-31-17-227 subjects]$ echo "LINUX STARTS" | tr 'A-Z' 'a-z'
linux starts
[rajkumari@ip-172-31-17-227 subjects]$ echo "linux stop" | tr 'a-z' 'A-Z'
LINUX STOP
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: sort

Objective -> To arrange the contents of a file in alphabetical or numerical order.

Purpose -> The sort command sorts text files line by line, making data easier to read and analyze.

```
[rajkumari@ip-172-31-17-227 subjects]$ cat file5.txt
rajkumari
linux
apachetomcat
devops
engineer
[rajkumari@ip-172-31-17-227 subjects]$ sort file5.txt
apachetomcat
devops
engineer
linux
rajkumari
[rajkumari@ip-172-31-17-227 subjects]$ s
```

Connecting to a server Commands

Cmd: su

Objective -> To switch from the current user account to another user account.

Purpose -> The su (substitute user) command allows a user to switch to another user, usually the **root** user, without logging out.

```
[rajkumari@ip-172-31-17-227 subjects]$ cd
[rajkumari@ip-172-31-17-227 ~]$ exit
exit
[root@ip-172-31-17-227 ~]# su rajkumari
[rajkumari@ip-172-31-17-227 root]$
```

Cmd: sudo

Objective -> To execute commands with administrative (root) privileges.

Purpose -> The sudo (Superuser Do) command allows authorized users to run commands as the root user without logging in as root.

```
[rajkumari@ip-172-31-17-227 ~]$ sudo yum update -y
Last metadata expiration check: 7:35:44 ago on Wed Jul  1 07:04:33 2026.
==================================================================================
WARNING:
  A newer release of "Amazon Linux" is available.

  Available Versions:

  Version 2023.12.20260629:
    Run the following command to upgrade to 2023.12.20260629:

      dnf upgrade --releasever=2023.12.20260629

    Release notes:
     https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.12.202606
29.html

==================================================================================
Dependencies resolved.
Nothing to do.
Complete!
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: ssh

Objective -> To securely connect to a remote Linux server over a network.

Purpose -> The ssh (Secure Shell) command is used to securely log in to a remote Linux system. It encrypts all communication between the client and the server, making it safe for remote administration, file management, and command execution.

Cmd: scp

Objective -> To securely copy files and directories between local and remote Linux systems.

Purpose -> The scp (Secure Copy) command transfers files over SSH, ensuring that data is encrypted during transmission.

Working with the Bash shell commands

Cmd: tee

Objective -> To display command output on the terminal and simultaneously save it to a file.

Purpose -> The tee command reads input from standard input (stdin), displays it on the screen, and writes it to one or more files at the same time.

```
[rajkumari@ip-172-31-17-227 subjects]$ ls
archive.zip  english    file2.txt  file4.txt  hindi.xz   physics  telugu
employ       file1.txt  file3.txt  file5.txt  marks.txt  social
[rajkumari@ip-172-31-17-227 subjects]$ ls | tee file5.txt
archive.zip
employ
english
file1.txt
file2.txt
file3.txt
file4.txt
file5.txt
hindi.xz
marks.txt
physics
social
telugu
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: history

Objective -> To display the list of previously executed commands in the current shell session.

Purpose -> The history command is used to view, manage, and reuse previously executed commands. It helps users avoid retyping commands and is useful for troubleshooting and auditing command history.

```
[rajkumari@ip-172-31-17-227 subjects]$ history
    1  whoami
    2  ls
    3  sudo ls
    4  su root
    5  cd
    6  exit
    7  sudo ls
    8  vi /etc/sudoers
    9  clear
   10  cd
   11  ls
   12  cd Desktop
   13  mkdir Desktop
   14  ls
   15  cd Desktop
   16  ls
   17  touch employee
   18  touch students
   19  college
   20  touchcollege
   21  touch college
   22  clear
   23  ls
   24  mkdir Techie
   25  mkdir horizo
   26  mv horizo horizon
   27  ls
   28  ip a
   29  ping google.com
```

```
 271  ls
 272  vi file5.txt
 273  cat file5.txt
 274  cut -c1-3 file5.txt
 275  cut -c4-8 file5.txt
 276  echo "LINUX STARTS" | tr 'a-z' 'A-Z'
 277  clear
 278  echo "LINUX STARTS" | tr 'A-Z' 'a-z'
 279  echo "linux stop" | tr 'a-z' 'A-Z'
 280  sort file4.txt
 281  sort file5.txt
 282  clear
 283  cat file5.txt
 284  sort file5.txt
 285  clear
 286  cd
 287  exit
 288  cd
 289  sudo apt update
 290  apt update
 291  sudo yum update -y
 292  clear
 293  cd DEsktop/subjects
 294  cd Desktop/subjects
 295  ls
 296  ls | tee file5.txt
 297  clear
 298  history
rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: history -d

Objective -> To delete a specific command from the command history.

Purpose -> The history -d command removes a selected command from the shell history. It is useful for deleting incorrect or sensitive commands.

```
[rajkumari@ip-172-31-17-227 subjects]$ history -d 292
[rajkumari@ip-172-31-17-227 subjects]$ history
```

```
 291  sudo yum update -y
 292  cd DEsktop/subjects
 293  cd Desktop/subjects
```

Cmd: alias

Objective -> To create a shortcut name for a Linux command.

Purpose -> The alias command allows users to define short names for long or frequently used commands, improving productivity and reducing typing.

```
[rajkumari@ip-172-31-17-227 subjects]$ alias ll='ls-l'
[rajkumari@ip-172-31-17-227 subjects]$ ll
bash: ls-l: command not found
[rajkumari@ip-172-31-17-227 subjects]$ ls -l
total 36
-rw-r--r--. 1 rajkumari rajkumari 162 Jul  1 08:00 archive.zip
-rw-rw-r--. 1 rajkumari rajkumari  70 Jul  1 14:16 employ
-rw-r--r--. 1 rajkumari rajkumari   0 Jul  1 07:57 english
-rw-rw-r--. 1 rajkumari rajkumari  22 Jul  1 10:03 file1.txt
-rw-rw-r--. 1 rajkumari rajkumari  21 Jul  1 10:07 file2.txt
-rw-rw-r--. 1 rajkumari rajkumari  33 Jul  1 11:01 file3.txt
-rw-rw-r--. 1 rajkumari rajkumari 795 Jul  1 11:49 file4.txt
-rw-rw-r--. 1 rajkumari rajkumari 118 Jul  1 14:44 file5.txt
-rw-r--r--. 1 rajkumari rajkumari  32 Jul  1 07:57 hindi.xz
-rw-rw-r--. 1 rajkumari rajkumari  41 Jul  1 12:16 marks.txt
-rw-r--r--. 1 rajkumari rajkumari   0 Jul  1 08:00 physics
-rw-r--r--. 1 rajkumari rajkumari   0 Jul  1 07:59 social
-rw-r--r--. 1 rajkumari rajkumari   0 Jul  1 07:56 telugu
```

User and Group Management and Permissions commands

Cmd: useradd

Objective -> To create a new user account in Linux.

Purpose -> The adduser command creates a new user, home directory, and default user settings, allowing the user to log in to the system.

```
[rajkumari@ip-172-31-17-227 subjects]$ sudo useradd Techie
[sudo] password for rajkumari:
[rajkumari@ip-172-31-17-227 subjects]$ cd
[rajkumari@ip-172-31-17-227 ~]$ cat /etc/passwd
```

```
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
rajkumari:x:1001:1001::/home/rajkumari:/bin/bash
Techie:x:1002:1002::/home/Techie:/bin/bash
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: usermod -L Techie

Objective: To lock an existing user account in Linux.

Purpose: The usermod -L command locks a user's password, preventing the user from logging into the system. It is commonly used to temporarily disable user access without deleting the account or the user's files.

```
[rajkumari@ip-172-31-17-227 ~]$ sudo usermod -L Techie
[sudo] password for rajkumari:
[rajkumari@ip-172-31-17-227 ~]$ sudo passwd -S Techie
Techie LK 2026-07-01 0 99999 7 -1 (Password locked.)
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: userdel Techie

Objective: To delete an existing user account from the Linux system.

Purpose: The userdel command removes a user account from the system. When used with the -r option, it also deletes the user's home directory and mail spool, making it useful for permanently removing unused user accounts.

```
[rajkumari@ip-172-31-17-227 ~]$ sudo userdel Techie
[rajkumari@ip-172-31-17-227 ~]$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
```

```
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
rajkumari:x:1001:1001::/home/rajkumari:/bin/bash
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: groupadd

Objective -> To display the group memberships of a user.

Purpose -> The groups command shows all groups to which a user belongs.

```
[rajkumari@ip-172-31-17-227 ~]$ sudo groupadd developers
[rajkumari@ip-172-31-17-227 ~]$ sudo groups
root
[rajkumari@ip-172-31-17-227 ~]$ groups developers
groups: 'developers': no such user
[rajkumari@ip-172-31-17-227 ~]$ sudo groups developers
[sudo] password for rajkumari:
groups: 'developers': no such user
[rajkumari@ip-172-31-17-227 ~]$ sudo grep developers /etc/group
developers:x:1002:
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: groupmod

Objective: To modify the properties of an existing group.

Purpose: The groupmod command is used to change an existing group's name or Group ID (GID). It helps administrators manage and update group information without affecting the users who belong to the group.

```
[root@ip-172-31-17-227 ~]# sudo groupmod -n Engineer developers
[root@ip-172-31-17-227 ~]# sudo grep Engineer /etc/group
Engineer:x:1002:
[root@ip-172-31-17-227 ~]#
```

Cmd: groudel

Objective: To delete an existing group from the Linux system.

Purpose: The groupdel command removes a group that is no longer required. Before deleting a group, ensure that it is not the primary group of any existing user

```
[rajkumari@ip-172-31-17-227 ~]$ sudo groupdel Engineer
[sudo] password for rajkumari:
[rajkumari@ip-172-31-17-227 ~]$ sudo grep developers /etc/group
[rajkumari@ip-172-31-17-227 ~]$ echo $?
1
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: id

Objective -> To display user and group identification information.

Purpose -> The id command displays the user's UID, GID, and group memberships

```
[rajkumari@ip-172-31-17-227 ~]$ id
uid=1001(rajkumari) gid=1001(rajkumari) groups=1001(rajkumari),10(wheel) context=un
confined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
```

Cmd: getent passwd rajkumari

Objective: To retrieve user account information from the system database.

Purpose: The getent passwd command displays user account details from local and network databases such as /etc/passwd, LDAP, or NIS. It is commonly used to verify user information, check account details, and troubleshoot authentication-related issues.

```
[rajkumari@ip-172-31-17-227 ~]$ getent passwd rajkumari
rajkumari:x:1001:1001::/home/rajkumari:/bin/bash
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: passwd rajkumari

Objective: To retrieve user account information from the system database.

Purpose: The getent passwd command displays user account details from local and network databases such as /etc/passwd, LDAP, or NIS. It is commonly used to verify user information, check account details, and troubleshoot authentication-related issues.

Cmd: chage -E 2026-12-31 rajkumari

Objective: To set an account expiration date for a user.

Purpose: The chage -E command is used to specify the date on which a user account will expire. After the expiration date, the user cannot log in until the account is re-enabled by the administrator.

Cmd: w

Objective: To display information about users currently logged into the system and their activities.

Purpose: The w command shows who is logged in, what they are doing, their login time, idle time, and system load. It is useful for monitoring user activity and overall system usage.

```
[rajkumari@ip-172-31-17-227 ~]$ w
 16:28:30 up 5 days,  6:38,  2 users,  load average: 0.00, 0.00, 0.00
USER     TTY        LOGIN@   IDLE   JCPU   PCPU WHAT
ec2-user pts/0     16:15    2.00s  0.02s  0.02s sshd: ec2-user [priv]
ec2-user pts/1     16:15    2.00s  0.04s  0.02s sudo -i
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: who

Objective: To display the users currently logged into the Linux system.

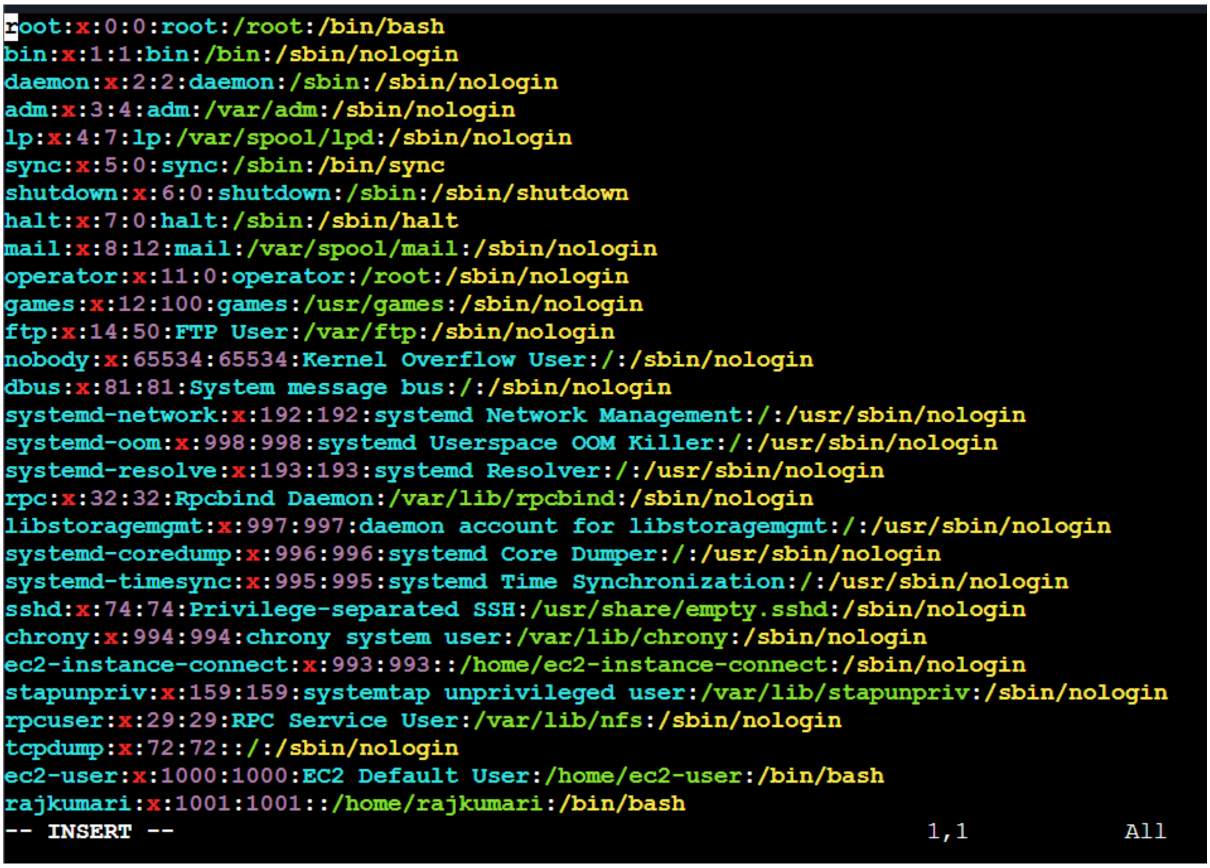
Purpose: The who command lists all currently logged-in users along with their terminal, login time, and remote host (if applicable). It is commonly used to check active user sessions and monitor system access.

```
[rajkumari@ip-172-31-17-227 ~]$ who
ec2-user pts/0        2026-07-01 16:15 (18.206.107.29)
ec2-user pts/1        2026-07-01 16:15 (18.206.107.29)
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: vipw

Objective: To safely edit the /etc/passwd file.

Purpose: The vipw command opens the /etc/passwd file in a text editor while locking it to prevent multiple users from editing it at the same time. This helps maintain the integrity of user account information and prevents file corruption.



```
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
rajkumari:x:1001:1001::/home/rajkumari:/bin/bash
-- INSERT --                                                  1,1           All
```

Cmd: vigr

Objective: To safely edit the /etc/group file.

Purpose: The vigr command opens the /etc/group file in a text editor while locking it to prevent simultaneous modifications. It is used to safely manage group information and avoid configuration errors.

```
root:x:0:
bin:x:1:
daemon:x:2:
sys:x:3:
adm:x:4:ec2-user
tty:x:5:
disk:x:6:
lp:x:7:
mem:x:8:
kmem:x:9:
wheel:x:10:ec2-user,rajkumari
cdrom:x:11:
mail:x:12:
man:x:15:
dialout:x:18:
floppy:x:19:
games:x:20:
tape:x:33:
video:x:39:
ftp:x:50:
lock:x:54:
audio:x:63:
users:x:100:
nobody:x:65534:
utmp:x:22:
utempter:x:35:
dbus:x:81:
ssh_keys:x:999:
input:x:104:
"/etc/group.edit" 52L, 769B
```

Cmd: loginctl

Objective: To manage and view user login sessions in systems using systemd.

Purpose: The loginctl command displays information about active user sessions, logged-in users, and system seats. It can also be used to terminate sessions, lock user sessions, and manage user login activities, making it useful for system administration and troubleshooting.

```
[rajkumari@ip-172-31-17-227 ~]$ login list-sessions
[rajkumari@ip-172-31-17-227 ~]$ echo $?
1
[rajkumari@ip-172-31-17-227 ~]$ which loginctl
/usr/bin/loginctl
[rajkumari@ip-172-31-17-227 ~]$ ps -p 1 -o comm=
systemd
[rajkumari@ip-172-31-17-227 ~]$ sudo systemctl status systemd-logind
[sudo] password for rajkumari:
● systemd-logind.service - User Login Management
     Loaded: loaded (/usr/lib/systemd/system/systemd-logind.service; static)
    Drop-In: /usr/lib/systemd/system/systemd-logind.service.d
             └─10-grub2-logind-service.conf
     Active: active (running) since Fri 2026-06-26 09:49:54 UTC; 5 days ago
       Docs: man:sd-login(3)
             man:systemd-logind.service(8)
             man:logind.conf(5)
```

Permissions Management Commands

Cmd: chown

Objective -> To change the ownership of a file or directory.

Purpose -> The chown command changes the owner and optionally the group of a file or directory.

```
[rajkumari@ip-172-31-17-227 ~]$ ls
Desktop  project.txt
[rajkumari@ip-172-31-17-227 ~]$ sudo chown Techie:developers project.txt
[rajkumari@ip-172-31-17-227 ~]$ ls -l project.txt
-rw-rw-r--. 1 Techie developers 0 Jul  1 16:58 project.txt
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: chgrp

Objective -> To change the group ownership of a file or directory.

Purpose -> The chgrp command assigns a different group to a file or directory while keeping the owner unchanged.

```
[rajkumari@ip-172-31-17-227 ~]$ sudo chgrp developers project.txt
[sudo] password for rajkumari:
[rajkumari@ip-172-31-17-227 ~]$ ls -l project.txt
-rw-rw-r--. 1 Techie developers 0 Jul  1 16:58 project.txt
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: chmod

Objective -> To modify file or directory permissions.

Purpose -> The chmod command changes read, write, and execute permissions to control access.

```
[rajkumari@ip-172-31-17-227 ~]$ sudo chmod 755 project.txt
[rajkumari@ip-172-31-17-227 ~]$ ls -l project.txt
-rwxr-xr-x. 1 Techie developers 0 Jul  1 16:58 project.txt
[rajkumari@ip-172-31-17-227 ~]$
```

Cmd: diff

Objective: To compare two files or directories and identify their differences.

Purpose: The diff command compares the contents of two files or directories line by line and displays the differences. It is commonly used to verify changes in configuration files, source code, and documents.

```
[rajkumari@ip-172-31-17-227 ~]$ cd Desktop/subjects
[rajkumari@ip-172-31-17-227 subjects]$ sudo diff file1.txt file2.txt
1,2c1
< Welcome to Blue Moon
<
---
> Hello everyone have
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: find

Objective -> To search for files and directories.

Purpose –> The find command searches for files and directories based on name, permissions, size, owner, or other criteria.

```
[rajkumari@ip-172-31-17-227 subjects]$ sudo find / -perm /4000
find: '/proc/205390/task/205390/fd/5': No such file or directory
find: '/proc/205390/task/205390/fdinfo/5': No such file or directory
find: '/proc/205390/fd/6': No such file or directory
find: '/proc/205390/fdinfo/6': No such file or directory
/usr/bin/sudo
/usr/bin/at
/usr/bin/chage
/usr/bin/gpasswd
/usr/bin/newgrp
/usr/bin/su
/usr/bin/mount
/usr/bin/umount
/usr/bin/staprun
/usr/bin/passwd
/usr/sbin/grub2-set-bootflag
/usr/sbin/pam_timestamp_check
/usr/sbin/unix_chkpwd
/usr/sbin/mount.nfs
```

Cmd: umask

Objective: To set or display the default file and directory permission mask.

Purpose: The umask command controls the default permissions assigned to newly created files and directories. It helps improve system security by restricting unnecessary access permissions.

```
[rajkumari@ip-172-31-17-227 subjects]$ umask
0002
[rajkumari@ip-172-31-17-227 subjects]$ umask 022
[rajkumari@ip-172-31-17-227 subjects]$ echo $?
0
[rajkumari@ip-172-31-17-227 subjects]$
```

Storage Management Essential commands

Cmd: lsblk

Objective: To display information about block storage devices.

Purpose: The lsblk command lists all available block devices such as hard disks, SSDs, partitions, and removable drives in a tree-like format. It helps administrators identify storage devices and their partition layouts.

```
[rajkumari@ip-172-31-17-227 subjects]$ lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: fdisk

Objective: To create, delete, modify, and manage disk partitions.

Purpose: The fdisk command is a disk partitioning utility used to view, create, resize, delete, and modify partitions on storage devices. It is commonly used during disk setup and storage management.

```
[rajkumari@ip-172-31-17-227 subjects]$ fdisk /dev/sda

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

fdisk: cannot open /dev/sda: No such file or directory
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: gdisk

Objective: To create and manage GPT (GUID Partition Table) disk partitions.

Purpose: The gdisk command is used to create, modify, delete, and manage GPT partitions on modern storage devices. It provides advanced partition management for disks larger than 2 TB and systems using the GPT partitioning scheme.

```
[rajkumari@ip-172-31-17-227 subjects]$ gdisk
GPT fdisk (gdisk) version 1.0.8

Type device filename, or press <Enter> to exit:
[rajkumari@ip-172-31-17-227 subjects]$ gdisk /dev/sda
GPT fdisk (gdisk) version 1.0.8

Problem opening /dev/sda for reading! Error is 2.
The specified file does not exist!
[rajkumari@ip-172-31-17-227 subjects]$
[rajkumari@ip-172-31-17-227 subjects]$ mkfs
mkfs: no device specified
Try 'mkfs --help' for more information.
[rajkumari@ip-172-31-17-227 subjects]$ mkfs.ext4 /dev/sda1
mke2fs 1.46.5 (30-Dec-2021)
The file /dev/sda1 does not exist and no size was specified.
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: mount

Objective: To mount a file system or storage device to the Linux directory structure.

Purpose: The mount command is used to attach a file system, partition, or storage device to a specific mount point, making its files and directories accessible to the operating system.

Cmd: findmnt

Objective: To mount a file system or storage device to the Linux directory structure.

Purpose: The mount command is used to attach a file system, partition, or storage device to a specific mount point, making its files and directories accessible to the operating system.

Cmd: umount

Objective: To safely unmount a mounted file system.

Purpose: The umount command detaches a mounted file system from the directory structure, ensuring that all pending data is written to disk before the device is removed or disconnected.

Cmd: lsof

Objective: To safely unmount a mounted file system.

Purpose: The umount command detaches a mounted file system from the directory structure, ensuring that all pending data is written to disk before the device is removed or disconnected.

Managing Networking commands

Cmd: ip a

Objective: To display network interface and IP address information.

Purpose: The ip a (or ip address) command shows all network interfaces, their IP addresses, MAC addresses, and interface status. It is commonly used to verify network configuration and troubleshoot connectivity issues

Cmd: ip route show

Objective: To display the system's routing table.

Purpose: The ip route show command displays all network routes configured on the system, including the default gateway and routes to different networks. It is useful for troubleshooting network routing and verifying connectivity settings.

```
[rajkumari@ip-172-31-17-227 subjects]$ ip route show
default via 172.31.16.1 dev ens5 proto dhcp src 172.31.17.227 metric 512
172.31.0.2 via 172.31.16.1 dev ens5 proto dhcp src 172.31.17.227 metric 512
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.17.227 metric 512
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.17.227 metric 512
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: ping

Objective: To test network connectivity between the local system and a remote host.

Purpose: The ping command sends ICMP Echo Request packets to a remote host and measures the response time. It is commonly used to verify network connectivity, check host availability, and troubleshoot network issues.

```
[rajkumari@ip-172-31-17-227 subjects]$ ping google.com
PING google.com (192.178.155.102) 56(84) bytes of data.
64 bytes from yuiadrs-in-f102.1e100.net (192.178.155.102): icmp_seq=1 ttl=106 time=
1.29 ms
64 bytes from yuiadrs-in-f102.1e100.net (192.178.155.102): icmp_seq=2 ttl=106 time=
1.36 ms
64 bytes from yuiadrs-in-f102.1e100.net (192.178.155.102): icmp_seq=3 ttl=106 time=
1.34 ms
64 bytes from yuiadrs-in-f102.1e100.net (192.178.155.102): icmp_seq=4 ttl=106 time=
1.34 ms
64 bytes from yuiadrs-in-f102.1e100.net (192.178.155.102): icmp_seq=5 ttl=106 time=
1.33 ms
64 bytes from yuiadrs-in-f102.1e100.net (192.178.155.102): icmp_seq=6 ttl=106 time=
1.34 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5009ms
rtt min/avg/max/mdev = 1.289/1.334/1.361/0.022 ms
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: ss

Objective: To display network socket and connection information.

Purpose: The ss command shows active TCP, UDP, and Unix socket connections, listening ports, and network statistics. It is commonly used to monitor network connections and troubleshoot networking problems.

```
[rajkumari@ip-172-31-17-227 subjects]$ ss -tuln
Netid State  Recv-Q Send-Q                  Local Address:Port  Peer Address:Port
udp   UNCONN 0      0                 172.31.17.227%ens5:68          0.0.0.0:*
udp   UNCONN 0      0                          127.0.0.1:323         0.0.0.0:*
udp   UNCONN 0      0      [fe80::8ff:e5ff:fee1:358f]%ens5:546          [::]:*
udp   UNCONN 0      0                              [::1]:323            [::]:*
tcp   LISTEN 0      128                          0.0.0.0:22          0.0.0.0:*
tcp   LISTEN 0      128                             [::]:22             [::]:*
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: dig

Objective: To query DNS servers for domain name information.

Purpose: The dig (Domain Information Groper) command is used to perform DNS lookups and retrieve information such as IP addresses, mail servers (MX records), name servers (NS records), and other DNS records. It is useful for DNS troubleshooting and verification.

```
[rajkumari@ip-172-31-17-227 subjects]$ dig example.com

; <<>> DiG 9.18.49 <<>> example.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 36057
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;example.com.                   IN      A

;; ANSWER SECTION:
example.com.            6       IN      A       172.66.147.243
example.com.            6       IN      A       104.20.23.154

;; Query time: 0 msec
;; SERVER: 172.31.0.2#53(172.31.0.2) (UDP)
;; WHEN: Wed Jul 01 17:38:22 UTC 2026
;; MSG SIZE  rcvd: 72

[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: nmap

Objective: To discover hosts and scan network ports.

Purpose: The nmap (Network Mapper) command is used to identify active hosts, detect open ports, discover running services, and assess network security. It is widely used for network administration, troubleshooting, and security auditing.

Cmd: hostname

Objective: To display or temporarily change the system's hostname.

Purpose: The hostname command displays the current hostname of the Linux system or temporarily changes it until the next reboot. It helps identify the system on a network.

```
[rajkumari@ip-172-31-17-227 subjects]$ hostname
ip-172-31-17-227.ec2.internal
[rajkumari@ip-172-31-17-227 subjects]$
```

Cmd: hostnamectl

Objective: To display and permanently manage the system hostname and related information.

Purpose: The hostnamectl command is used to view and modify the system's static, transient, and pretty hostnames on systems using systemd. It also displays operating system and kernel information, making it useful for system identification and administration.

Working with Systemd commands

Cmd: systemctl start

Example: systemctl start httpd
Description: Starts a specified unit file.
Cmd: systemctl status

Example: systemctl status httpd
Description: Displays the current status of a unit file.
Cmd: systemctl restart

Example: systemctl restart httpd
Description: Restarts a unit file, often used after making configuration changes.
Cmd: systemctl stop

Example: systemctl stop httpd
Description: Stops a running unit file.
Cmd: systemctl enable

Example: systemctl enable httpd
Description: Enables a unit file to start automatically on system boot.
Cmd: systemctl disable

Example: systemctl disable httpd
Description: Disables a unit file from starting

Cmd:systemctl list-units

Example: systemctl list-units
Description: Lists all currently loaded unit files.

Cmd: systemctl set-default

Example: systemctl set-default graphical.target

Description: Sets the default target that the system boots into.

Cmd: systemctl get-default

Example: systemctl get-default

Description: Displays the current default target.

Cmd: systemctl cat

Example: systemctl cat httpd

Description: Shows the configuration of a unit file.

Cmd: systemctl show

Example: systemctl show httpd

Description: Displays all the options set for a unit file.

Cmd: systemctl edit

Example: systemctl edit httpd

Description: Opens an editor to modify a unit file.

Cmd: systemctl daemon-reload

Example: systemctl daemon-reload

Description: Instructs the systemctl main process to reload its configuration.

Cmd: systemctl isolate

Example: systemctl isolate graphical.target

Description: Switches the system to a different target.

Cmd: systemctl list-dependencies

Example: systemctl list-dependencies graphical.target

Description: Lists all dependencies of a specific target.

Managing Software Commands

Command: rpm -> Description: Originally used for installing packages, rpmis now mainly used for querying information about installed packages.

Example: rpm -q package_name - This command queries details about a specific package.

Command: yum -> Description: yumis a software package manager used in Red Hat-based environments for installing, updating, and managing packages.

Example: yum install package_name - This command installs a specific package.

Command: dnf -> Description: Similar to yum, dnfis a modern package manager for Red Hat-based systems, offering improved performance and package management capabilities.

Example: dnf update- This command updates all installed packages to their latest versions.

Command: apt -> Description: aptis a command-line tool used in Ubuntu and other Debian-based systems for handling packages, including installation, updating, and removal.

Example: apt install package_name - This command installs a specific package.

Command: apt-cache -> Description: apt-cacheis used for querying package information in Debian-based systems. It's especially useful for searching for packages containing specific files.

Example: apt-cache search keyword - This command searches for packages related to a specific keyword

Managing SSH Commands

Change SSH Port on Ubuntu:

Command: **sudo vim /etc/ssh/sshd_config**

Description: Open the SSH configuration file using vimto change the SSH port.

Update SSH Port to 2022:

Command: Change the line Port 22to Port 2022in the file.

Description: In the SSH configuration file, update the default port from 22 to 2022.

Restart SSH Service:

Command: **sudo systemctl restart sshd**

Description: Restart the SSH daemon to apply the new configuration.

Allow Port 2022 in Firewall:

Command: sudo ufw allow 2022/tcp

Description: Update the Ubuntu firewall to allow TCP connections on port 2022.

Check SSH Service Status:

Command: sudo systemctl status sshd

Description: Verify that the SSH service is active and listening on port 2022.

Create SSH Session from Windows:

Description: Use MobaXterm on Windows to create a new SSH session.

Steps:

Open MobaXterm and select to create a new SSH session.

Enter the remote host IP address (e.g., 192.168.29.142).

Specify the username (studentin this case).

Set the port to 2022.

Click OK to initiate the connection, then enter the password when prompted

Managing Time commands

Command: date -> Description: Displays the current date and time of the system. It's a basic and long-standing command in many operating systems.

Example: date

```
[rajkumari@ip-172-31-17-227 subjects]$ date
Wed Jul  1 18:08:41 UTC 2026
[rajkumari@ip-172-31-17-227 subjects]$
```

Command: hwclock -> Description: Used for managing and viewing hardware clock. It can also be used to synchronize the hardware clock with the system clock.

Example: hwclock –show

```
[rajkumari@ip-172-31-17-227 subjects]$ hwclock --show
hwclock: Cannot access the Hardware Clock via any known method.
hwclock: Use the --verbose option to see the details of our search for an access me
thod.
[rajkumari@ip-172-31-17-227 subjects]$
```

Command: timedatectl -> Description: A modern utility for managing various aspects of time synchronization and settings. It can display and set time, date, and timezone, among other features.

Example: timedatectl status

```
[rajkumari@ip-172-31-17-227 subjects]$ timedatectl status
               Local time: Wed 2026-07-01 18:10:41 UTC
           Universal time: Wed 2026-07-01 18:10:41 UTC
                 RTC time: Wed 2026-07-01 18:10:40
                Time zone: n/a (UTC, +0000)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
[rajkumari@ip-172-31-17-227 subjects]$
```

Command: chronyc -> Description: Interfaces with the chronydtime-synchronizing service, providing detailed information about the current synchronization status and more control over the chronyd service.

Example: chronyc tracking

```
[rajkumari@ip-172-31-17-227 subjects]$ chronyc tracking
Reference ID    : A9FEA97B (169.254.169.123)
Stratum         : 4
Ref time (UTC)  : Wed Jul 01 18:11:35 2026
System time     : 0.000000790 seconds fast of NTP time
Last offset     : +0.000000474 seconds
RMS offset      : 0.000006709 seconds
Frequency       : 7.089 ppm slow
Residual freq   : +0.000 ppm
Skew            : 0.153 ppm
Root delay      : 0.000405049 seconds
Root dispersion : 0.000265048 seconds
Update interval : 16.2 seconds
Leap status     : Normal
[rajkumari@ip-172-31-17-227 subjects]$
```

Process Management Commands

Command: top -> Description: Monitors all processes and the general health of the system from a performance perspective.

Example: top

```
top - 18:18:25 up 5 days,  8:28,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 107 total,   1 running, 106 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    912.9 total,    209.7 free,    221.1 used,    482.0 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    512.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1062 root      20   0       0      0      0 S   0.3   0.0   1:50.92 xfsaild/nv+
      1 root      20   0  173220  17700  10976 S   0.0   1.9   0:23.40 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.09 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workq+
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-+
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-+
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-+
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:+
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-+
     14 root      20   0       0      0      0 S   0.0   0.0   0:01.36 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:04.72 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_pa+
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.18 rcu_exp_gp+
     18 root      rt   0       0      0      0 S   0.0   0.0   0:01.84 migration/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
```

Command: ps -> Description: Shows all processes and their properties; useful for monitoring system processes.

Example: ps aux

```
[rajkumari@ip-172-31-17-227 subjects]$ ps
    PID TTY          TIME CMD
 203669 pts/1    00:00:00 bash
 207164 pts/1    00:00:00 ps
[rajkumari@ip-172-31-17-227 subjects]$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  1.8 173220 17700 ?        Ss   Jun26   0:23 /usr/lib/systemd
root           2  0.0  0.0      0     0 ?        S    Jun26   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    Jun26   0:00 [pool_workqueue_
root           4  0.0  0.0      0     0 ?        I<   Jun26   0:00 [kworker/R-rcu_g
root           5  0.0  0.0      0     0 ?        I<   Jun26   0:00 [kworker/R-sync_
root           6  0.0  0.0      0     0 ?        I<   Jun26   0:00 [kworker/R-kvfre
root           7  0.0  0.0      0     0 ?        I<   Jun26   0:00 [kworker/R-slub_
root           8  0.0  0.0      0     0 ?        I<   Jun26   0:00 [kworker/R-netns
root          10  0.0  0.0      0     0 ?        I<   Jun26   0:00 [kworker/0:0H-ev
root          13  0.0  0.0      0     0 ?        I<   Jun26   0:00 [kworker/R-mm_pe
root          14  0.0  0.0      0     0 ?        S    Jun26   0:01 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        I    Jun26   0:04 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    Jun26   0:00 [rcu_exp_par_gp_
root          17  0.0  0.0      0     0 ?        S    Jun26   0:00 [rcu_exp_gp_kthr
root          18  0.0  0.0      0     0 ?        S    Jun26   0:01 [migration/0]
root          19  0.0  0.0      0     0 ?        S    Jun26   0:00 [cpuhp/0]
```

Command: jobs -> Description: Lists the jobs (processes) started by the user in the shell, allowing for management of interactive processes.

Example: jobs

```
[rajkumari@ip-172-31-17-227 subjects]$ jobs
[rajkumari@ip-172-31-17-227 subjects]$ echo $?
0
[rajkumari@ip-172-31-17-227 subjects]$
```

Command: fg -> Description: Brings a job to the foreground, where %1is the job number.

Example: fg %1

Command: bg -> Description: Sends a job to run in the background.

Example: bg %1

Command: nice -> Description: Starts a process with a defined niceness (priority), where commandis the command to be run with a priority.

Example: nice -n 10 command

```
[rajkumari@ip-172-31-17-227 root]$ nice -n 10 command
[rajkumari@ip-172-31-17-227 root]$ echo $?
0
[rajkumari@ip-172-31-17-227 root]$ 
```

Command: renice -> Description: Changes the priority of an already running process, where 1234is the Process ID (PID).

Example: renice -n 10 -p 1234

Command: kill ->Description: Sends a signal to a process (usually to terminate it), where 1234is the Process ID (PID).

Command: killall -> Description: Terminates all processes with the given name.

Example: killall processname

Scheduling Tasks Commands

Command: at -> Description: Allows you to schedule a job to run at a specific time.

Example: at 10:30schedules a job to run at 10:30.

Command: atg -> Description: Displays a list of all jobs currently scheduled.

Example: atqshows all scheduled jobs with their job numbers.

Command: atrm -> Description: Used to remove scheduled jobs.

Example: atrm 5removes the job with the job number 5.

```
[rajkumari@ip-172-31-17-227 ~]$ atrm
Usage: at [-V] [-q x] [-f file] [-mMlbv] timespec ...
       at [-V] [-q x] [-f file] [-mMlbv] -t time
       at -c job ...
       atq [-V] [-q x]
       at [ -rd ] job ...
       atrm [-V] job ...
       batch
[rajkumari@ip-172-31-17-227 ~]$ crontab -e
bash: crontab: command not found
[rajkumari@ip-172-31-17-227 ~]$
```

Editing Cron Configuration with command: crontab -e

Description: Opens the cron configuration for editing, allowing you to schedule jobs to run periodically.

Example: crontab -eopens the cron configuration file in a text editor.

Working with Systemctl Timers Using: systemctl

Description: Used to manage and interact with systemctl timers, a more systemwide approach for scheduling tasks.

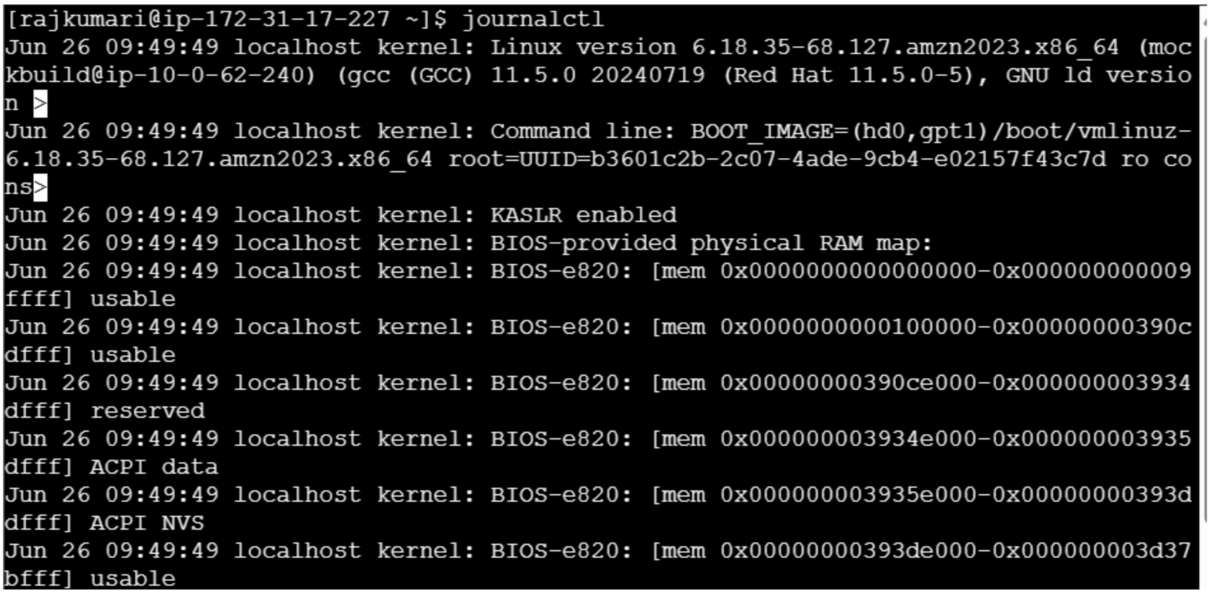
Example: **systemctl list-timers**shows all active timers.

```
[rajkumari@ip-172-31-17-227 ~]$ systemctl
  UNIT                                                                       LOAD
   ACTIVE SUB       DESCRIPTION
  boot-efi.automount                                                         load
ed active running   boot-efi.automount
  proc-sys-fs-binfmt_misc.automount                                          load
ed active running   Arbitrary Executable File Formats File System Automount Point
  sys-devices-pci0000:00-0000:00:04.0-nvme-nvme0-nvme0n1-nvme0n1p1.device      load
ed active plugged   Amazon Elastic Block Store /
  sys-devices-pci0000:00-0000:00:04.0-nvme-nvme0-nvme0n1-nvme0n1p127.device    load
ed active plugged   Amazon Elastic Block Store BIOS\x20Boot\x20Partition
  sys-devices-pci0000:00-0000:00:04.0-nvme-nvme0-nvme0n1-nvme0n1p128.device    load
ed active plugged   Amazon Elastic Block Store EFI\x20System\x20Partition
  sys-devices-pci0000:00-0000:00:04.0-nvme-nvme0-nvme0n1.device              load
ed active plugged   Amazon Elastic Block Store
  sys-devices-pci0000:00-0000:00:05.0-net-ens5.device                        load
ed active plugged   Elastic Network Adapter (ENA)
  sys-devices-platform-serial8250-serial8250:0-serial8250:0.1-tty-ttyS1.device load
ed active plugged   /sys/devices/platform/serial8250/serial8250:0/serial8250:0.1/tt
```

Reading Log Files Commands

Command: journalctl -> Description: This command is used to check the contents of the systemd journal, which is a centralized logging system for Linux.

Example: journalctl -u nginx.service
This example shows how to view logs for the Nginx service using journalctl.



Command: tail -> Description: The tailcommand is used to view the contents of log files, typically the last few lines.

Example: tail -n 100 /var/log/syslog
This example displays the last 100 lines of the syslog file.
Command: less -> Description: Similar to tail, lessis used for viewing log files but allows for more controlled navigation through the file.

Example: less /var/log/apache2/access.log
This opens the Apache access log file in a scrollable interface for in-depth examination.

Command: logger -> Description: The loggercommand is a convenient tool for writing messages to the system log.

Example: logger "System backup completed"
This example sends a custom message, "System backup completed," to the system log.

```
[rajkumari@ip-172-31-17-227 ~]$ logger "system backup completed"
[rajkumari@ip-172-31-17-227 ~]$ echo $?
0
[rajkumari@ip-172-31-17-227 ~]$
```